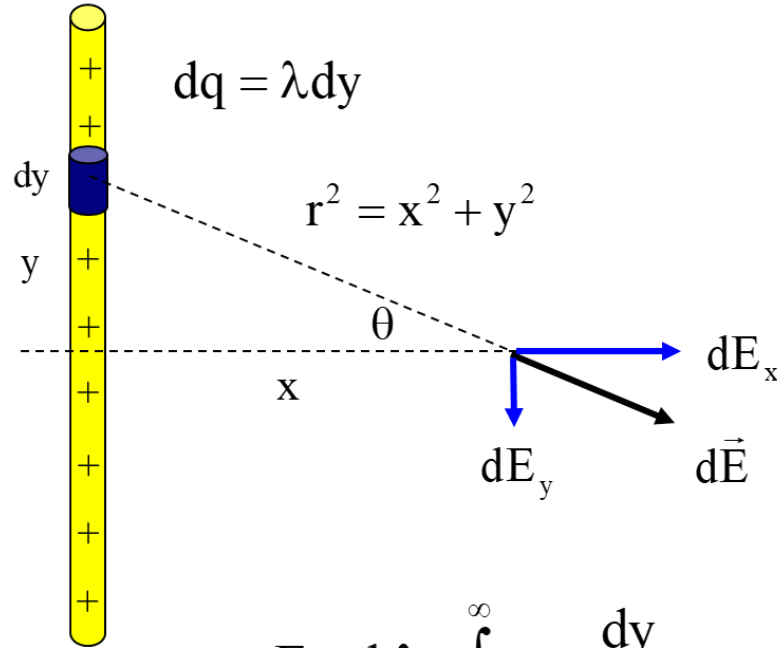


Extended: Infinitely Long Line of Charge



$$d\vec{E} = \frac{k dq}{r^2} \hat{r}$$

y-components cancel by symmetry

$$dE_x = \frac{k dq}{r^2} \cos \theta$$

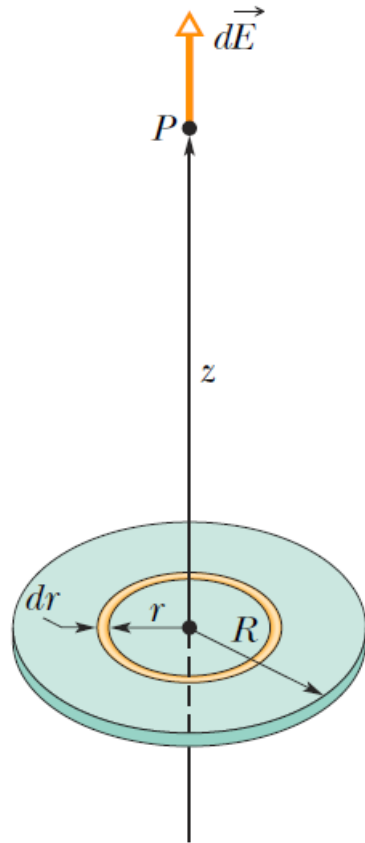
$$dE = \frac{k(\lambda dy)}{x^2 + y^2} \frac{x}{\sqrt{x^2 + y^2}}$$

$$E = k\lambda x \int_{-\infty}^{\infty} \frac{dy}{(x^2 + y^2)^{\frac{3}{2}}} = k\lambda x \left(\frac{2}{x^2} \right) = \frac{2k\lambda}{x}$$

$$\int \frac{dx}{(x^2 + a^2)^{3/2}} = \frac{x}{a^2(x^2 + a^2)^{1/2}}$$

Inversely proportional to the distance!

Extended: Infinite Charged Sheet



When $R \rightarrow \infty$,

$$E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{z^2 + R^2}} \right] \approx \frac{\sigma}{2\epsilon_0}$$

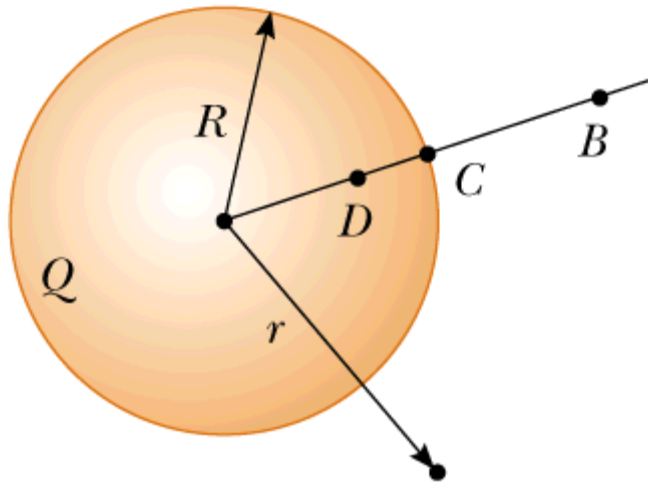
Why E is independent of the distance z?

Translational (planar) symmetry!

Similar to the **Square** problem when **Z** to zero

Recap of gravitation: Uniformly Charged Solid Sphere

Problem: Find the electric field of a uniformly charged solid sphere at B, C, and D. Parameters are shown in figure.



At B, the electric field should be the same as if there is a point charge Q at the sphere centre. Similar to how we compute gravity.

Newton got the above result for gravity, but spent a very long time to prove this as he has no calculus!

Then how about D?

We need another great scientist to tell the answer.

Gauss' Law



Carl Friedrich Gauss

Gauss was a German mathematician who contributed significantly to many fields, including number theory, algebra, statistics, differential geometry, geophysics, electrostatics, astronomy, etc.

Puzzle of his birthday

Gauss' mother could not remember his birthday. All she remember was that it was a Wednesday, eight days before Ascension Day, which occurs 40 days after Easter Sunday. 22 years later, Gauss himself found a way to determine the day of Easter for all years. From this he found his own birthday to be April 30, 1777.



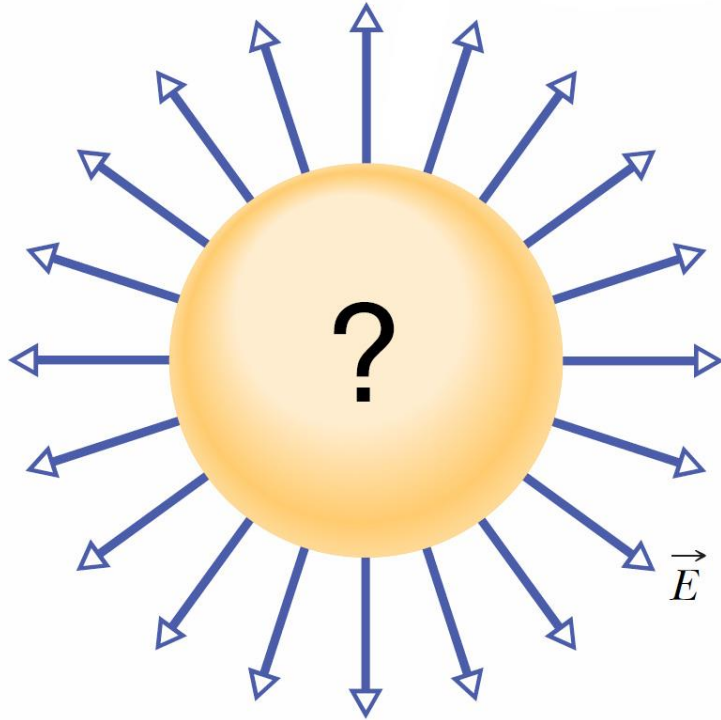
Gauss (1777 – 1855)

http://american_almanac.tripod.com/gauss.htm

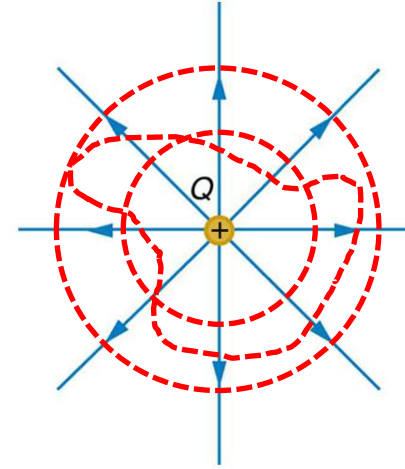
Today's topics

- Electric Flux – Pre Gauss's law
- Gauss' Law
- Electric Shielding - Metal
- Various Symmetries - Gauss's law applied

A “Simple” Question



How much electric charge is there?



$$|\vec{E}| = \frac{kQ}{r^2}$$

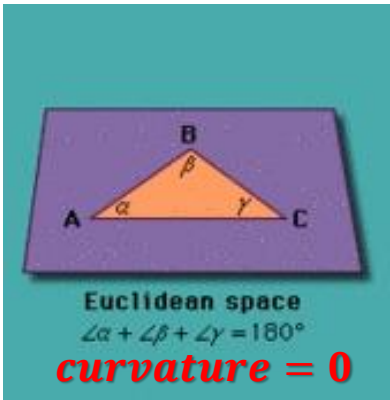
For spheres: $|\vec{E}| \cdot S = \frac{kQ}{r^2} \cdot 4\pi r^2 = \text{const.}$

Gauss' Law

The net electric **flux** Φ through a closed surface (called a “Gaussian surface”) is proportional to the net charge *enclosed* by that surface

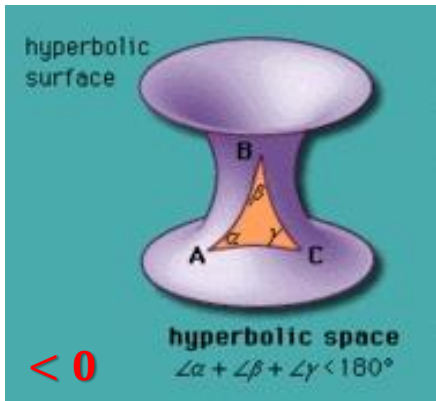
Non-Euclidean Geometry Pioneered by Gauss*

欧氏几何



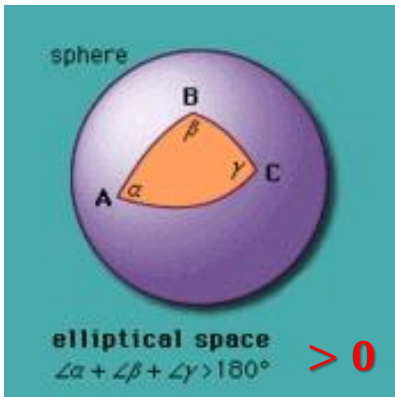
Euclidean geometry

罗氏几何

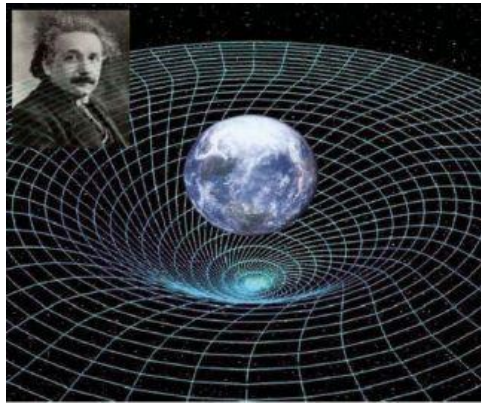


Lobachevsky geometry

黎曼几何



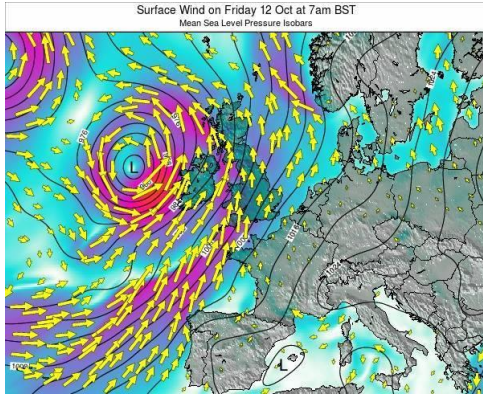
Riemann geometry
(Graduate of Gauss)



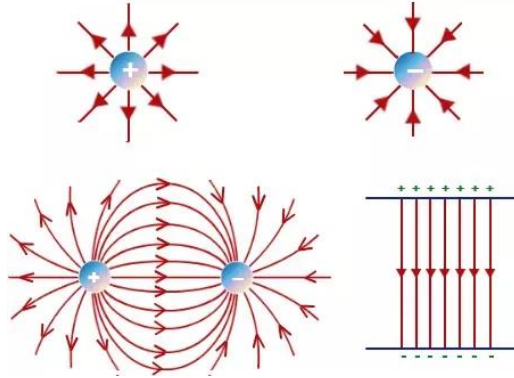
Theory of Relativity is established based on the Non-Euclidean space.

<https://baijiahao.baidu.com/s?id=1672681188359610030&wfr=spider&for=pc>

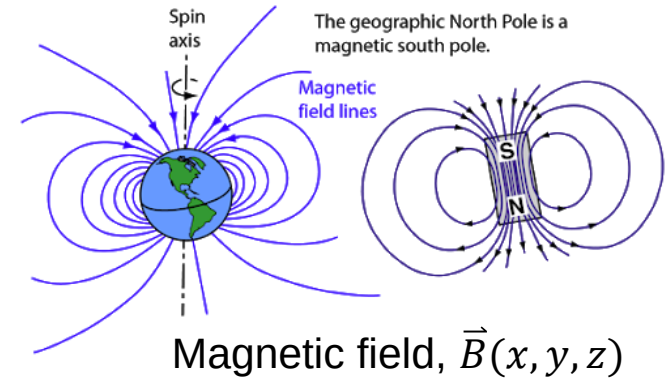
Vector Field



Wind/fluid field



Electric field, $\vec{E}(x, y, z)$



■ Vector **f**ield can often be described by a special function of $\vec{f}(x, y, z)$.

■ The vector field can be schematically illustrated with **field lines** (direction, density)

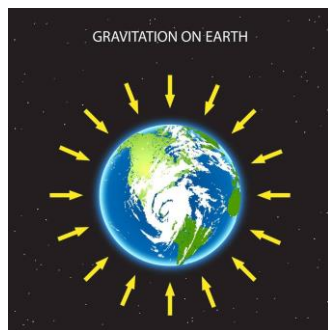
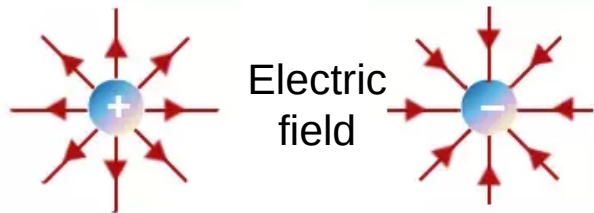


Gravitational field, $\vec{G}(x, y, z)$

Field With / Without Sources

With sources (有源场)

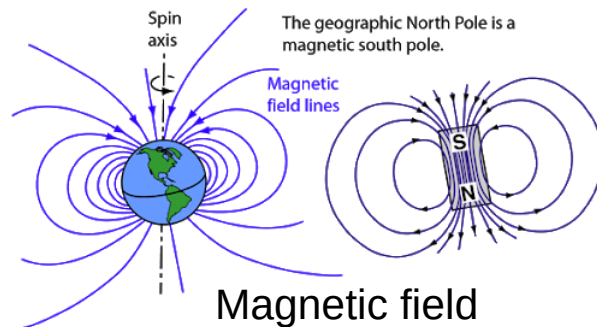
Divergent field (发散场)



Gravitational field
on earth

Without sources (无源场)

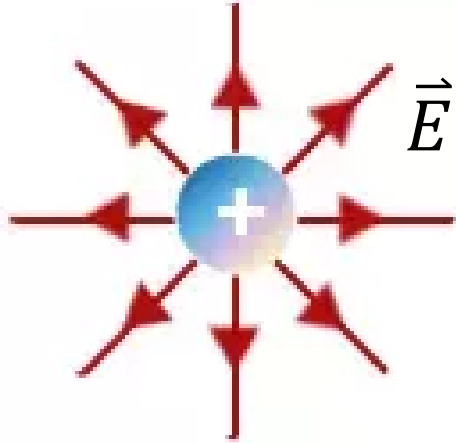
Curl field (有旋场)



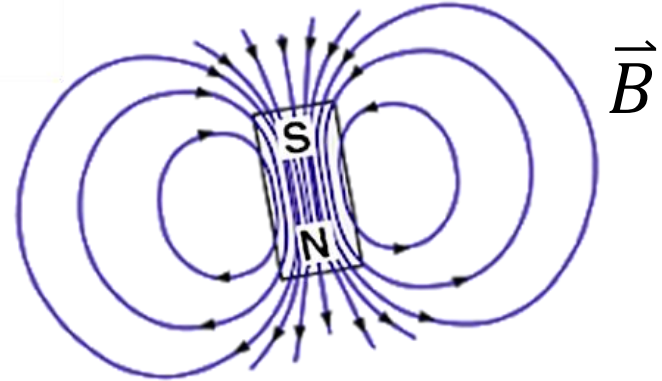
Fluid field



Divergence vs. curl of field



Electric field / divergent field



Magnetic field / curl field

Divergence : $\vec{\nabla} \cdot \vec{E} \neq 0$

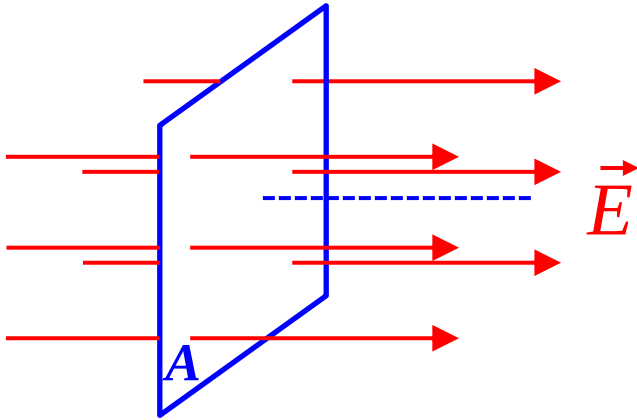
Curl : $\vec{\nabla} \times \vec{E} = 0$

$\vec{\nabla} \cdot \vec{B} = 0$

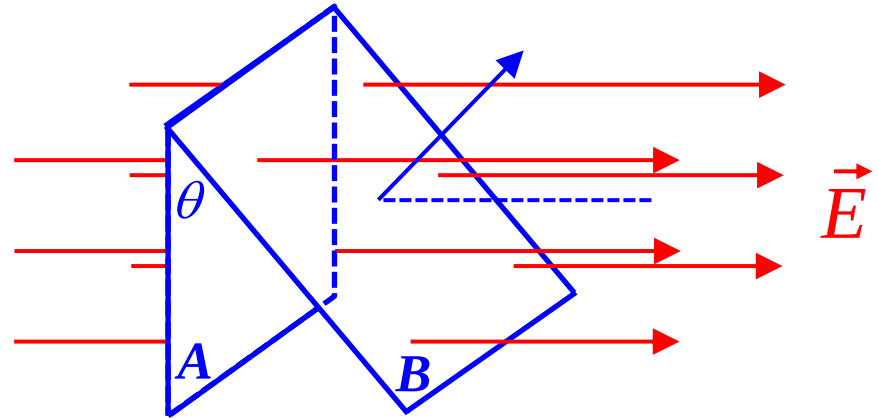
$\vec{\nabla} \times \vec{B} \neq 0$

Electric Flux

- * **Electric flux Φ** is defined by the amount of electric field $\mathbf{E}$ *piercing* an area $\mathbf{A}$.



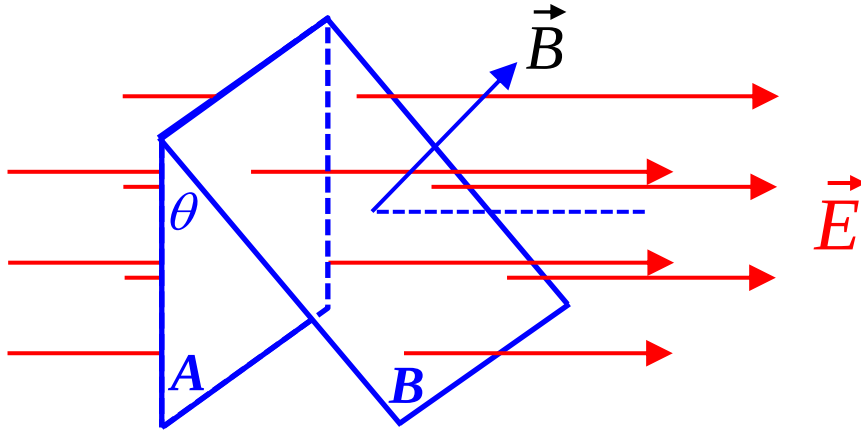
$$\Phi_A = S_A \cdot |\vec{E}|$$



~~$$\Phi_B = S_B \cdot |\vec{E}|?$$~~

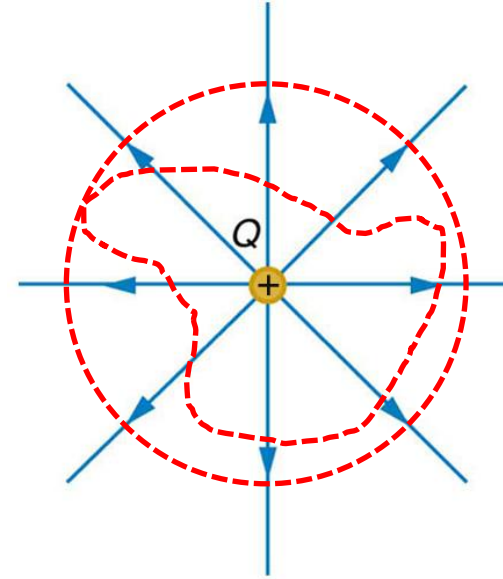
$$\Phi_B = S_A \cdot |\vec{E}| = S_B \cos \theta \cdot |\vec{E}|$$

Surface Vector



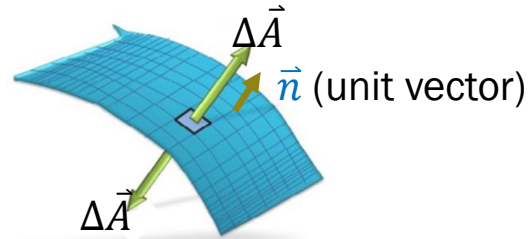
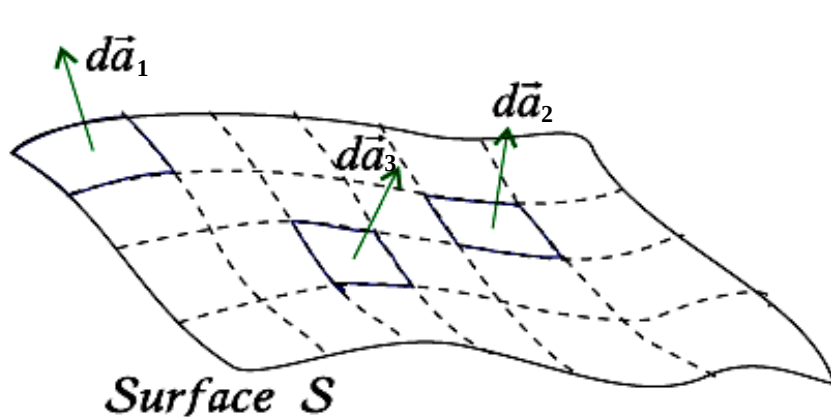
$$\begin{aligned}\Phi_B &= S_A \cdot |\vec{E}| = S_B \cos \theta \cdot |\vec{E}| \\ &= \vec{B} \cdot \vec{E}\end{aligned}$$

Define the surface vector $\vec{B}$: its magnitude is its area, and its direction is along the normal.



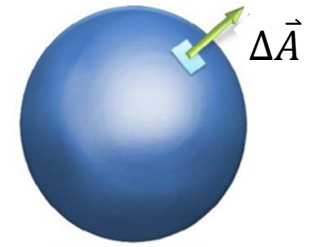
How to define surface vector on a sphere?
What about a closed surface of arbitrary shape?

Surface Vector



Open surface

Any one of the two normal as the outward normal

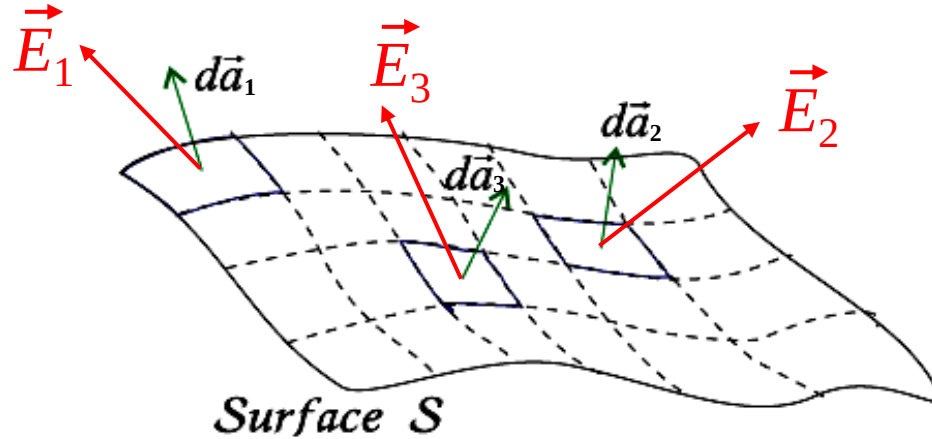


Closed surface

Only one possible outward normal

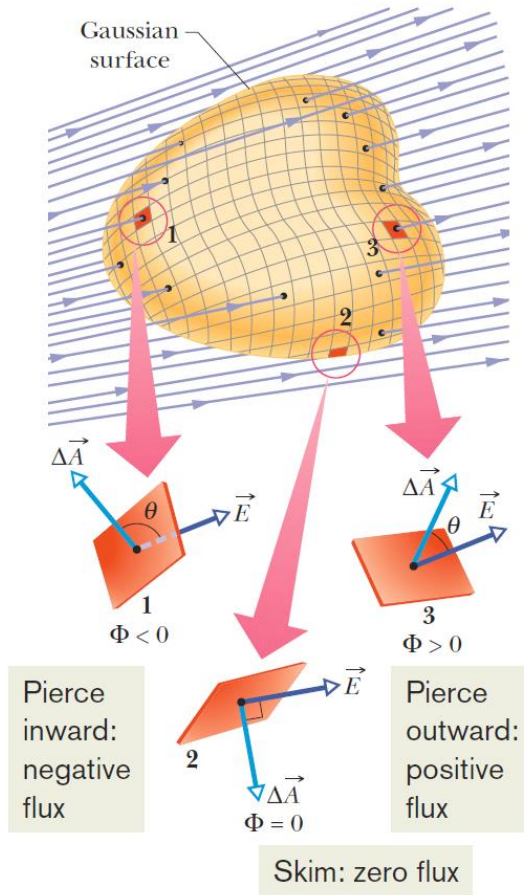
Apply the idea of differential: dividing an arbitrary surface into many tiny patches, so each patch can be approximately viewed as a small plane whose normal is well defined. $d\vec{a}_i$ is a vector that is always perpendicular to the surface S with infinitesimal area da_i .

Electric Flux through Arbitrary Surface



$$\Phi = \sum \Delta\Phi = \sum \vec{E}_i \cdot \Delta\vec{a}_i \rightarrow \int d\Phi = \int \vec{E} \cdot d\vec{a}$$

Electric Flux



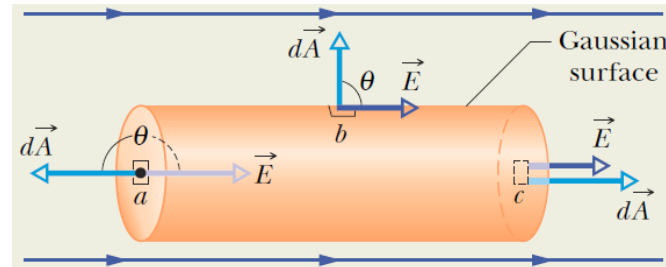
- The net flux for any extended surface is

$$\Phi = \int_{\mathbf{A}} \mathbf{E} \cdot d\mathbf{A}$$

- For a **closed** surface,

$$\Phi = \oint \mathbf{E} \cdot d\mathbf{A}$$

Example: The net flux is zero for a cylindrical surface placing in a uniform E -field. (**Why?**)



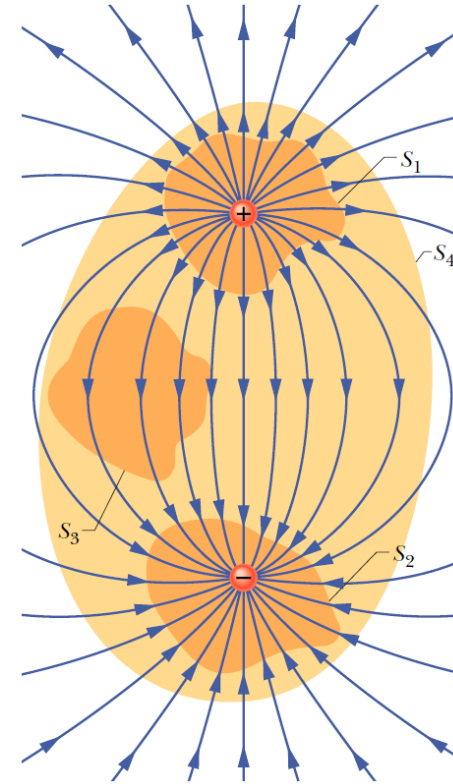
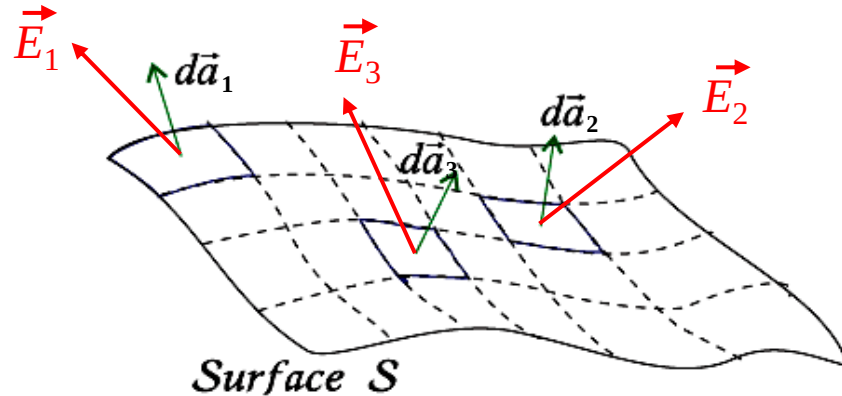
Today's topics

- Electric Flux
- Gauss' Law
- Electric Shielding
- Various Symmetries

Gauss' Law

The net electric flux Φ through a closed surface (called a “Gaussian surface”) is proportional to the net charge *enclosed* by that surface

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{enc}}}{\epsilon_0}$$

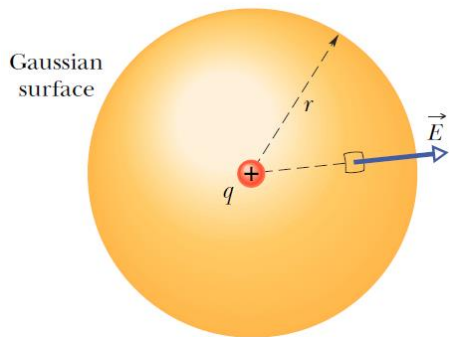


$$\Phi_1 = q/\epsilon_0, \Phi_2 = -q/\epsilon_0, \Phi_3 = 0, \\ \Phi_4 = +q/\epsilon_0 - q/\epsilon_0 = 0.$$

Gauss's Law is Equivalent to Coulomb's Law

1. Gauss' Law $\Rightarrow$ Coulomb's Law

Consider a system with only one isolated charge q . Construct a spherical Gaussian surface centered at the charge (see figure). In this case, $\mathbf{E}$ is *constant* and *parallel* to $d\mathbf{A}$ anywhere on the surface, so



$$\mathbf{E} \cdot d\mathbf{A} = E dA.$$

Therefore, according to Gauss' law,

$$\frac{q}{\epsilon_0} = \oint \mathbf{E} \cdot d\mathbf{A} = \oint E dA = E \oint dA = 4\pi r^2 E$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

2. Coulomb's Law $\Rightarrow$ Gauss' Law (Try it yourself!)

What is Gauss' Law?

- Gauss' Law **does not tell us anything new**, it is NOT a new law of physics, but another way of expressing Coulomb's Law
- Gauss' law makes it possible to find the electric field easily in **highly symmetric situations**.
- The differential form of Gauss' law is the same as **Maxwell's first equation**, and both are equivalent to Coulomb's law.

Gauss' Law is Maxwell's First Equation

$$\text{Gauss' law: } \oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{enc}}}{\epsilon_0}$$

$$\text{Introduce the Nabla symbol } \nabla \equiv \frac{\partial}{\partial x} \mathbf{e}_x + \frac{\partial}{\partial y} \mathbf{e}_y + \frac{\partial}{\partial z} \mathbf{e}_z$$

The mathematical **divergence theorem (Gauss' theorem)** states that the outward flux of a vector field $\mathbf{F}$ through a closed surface is equal to the volume integral of the **divergence** over the region inside the surface:

$$\int_V \nabla \cdot \mathbf{F} dV = \oint_{\mathbf{A}=\partial V} \mathbf{F} \cdot d\mathbf{A}$$

Apply this theorem to the left of the Gauss' law, and remember that q_{enc} equals to the integral of ρ over the volume of the Gaussian surface, we get

$$\oint_{\mathbf{A}} \mathbf{E} \cdot d\mathbf{A} = \int_V \nabla \cdot \mathbf{E} dV = \frac{1}{\epsilon_0} \int_V \rho dV = \frac{q_{\text{enc}}}{\epsilon_0}$$

Getting rid of the integral yields the differential form of the Gauss' law:

$$\nabla \cdot \mathbf{E} = \rho / \epsilon_0$$

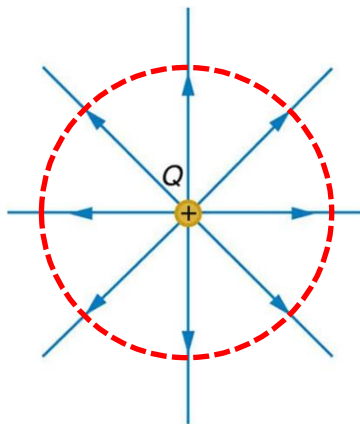
So the differential form of Gauss' law is the same as Maxwell's first equation, and both are equivalent to Coulomb's law.

Brain Teasers

A balloon has a single point charge in its interior. Does the electric flux through the balloon depend on whether or not it is fully inflated? Explain your reasoning.



A spherical Gaussian surface encloses a point charge q . If the point charge is moved from the center of the sphere to a point away from the center, does the electric field at a point on the surface change? Does the total flux through the Gaussian surface change?



- (a) In a certain region of space, the volume charge density ρ has a uniform positive value. Can E be uniform in this region? Explain.
- (b) Suppose that in this region of uniform positive ρ there is a “bubble” within which $\rho = 0$. Can E be uniform within this bubble? Explain.

Today's topics

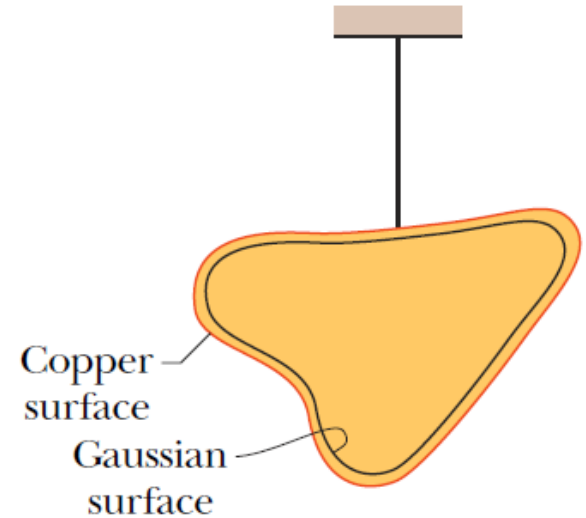
- Electric Flux – Pre Gauss's law
- Gauss' Law
- Electric Shielding - Metal
- Various Symmetries - Gauss's law applied

Applying Gauss' Law – Charged Isolated Conductor

Theorem: All excess charges in a charged isolated conductor is placed on the *outer surface* of the conductor.

Explain:

1. **In electrostatic equilibrium (all charges stable), electric field is zero for all points inside the conductor.** If not, electric forces would move the charges.
2. Construct the Gaussian surface as in figure. Since $E = 0$ for all points at the surface, $\Phi = 0$. According to Gauss' law, net charge $q = 0$ inside the Gaussian surface.



Therefore, excess charges must be on the outer surface.

E-field of a charged isolated **conducting** sphere

Sketch the E-field against the distance r .

Remember the Gauss's law:

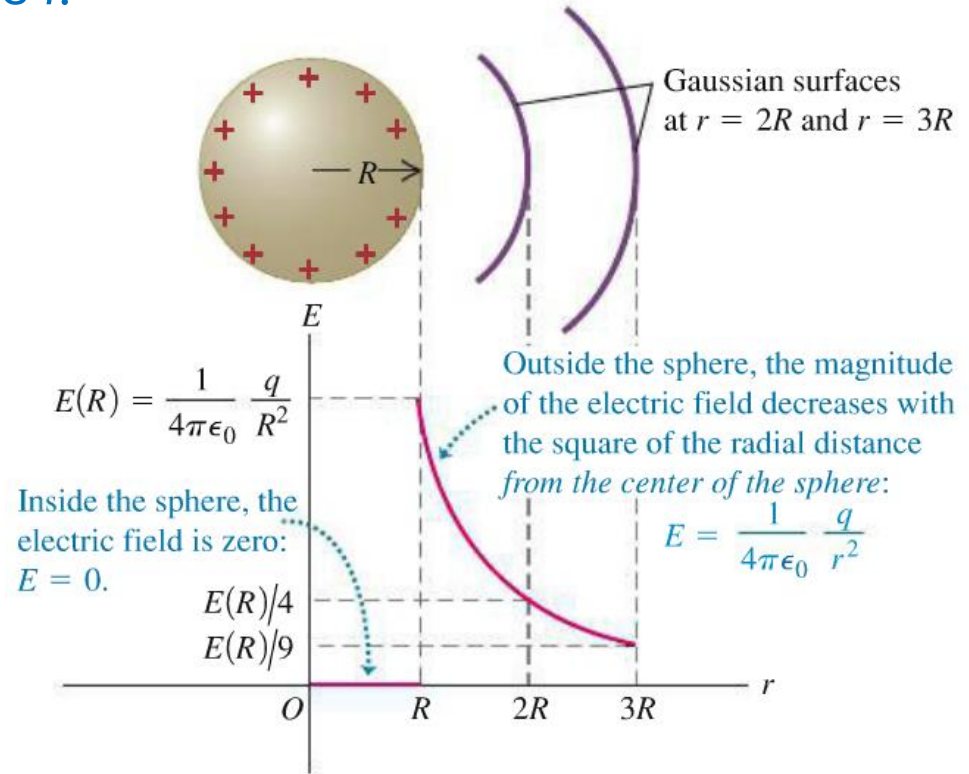
$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$

Outside of sphere, $r \geq R$:

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

Inside of sphere, $r < R$:

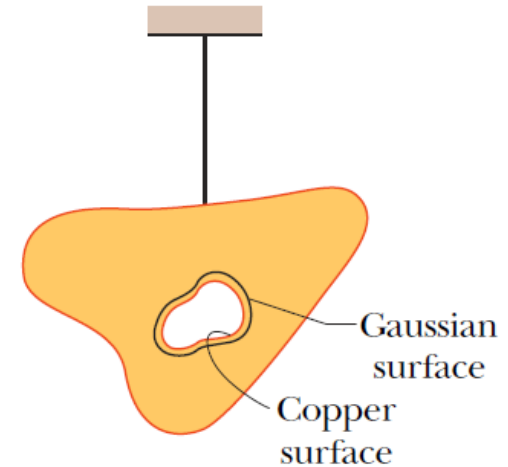
Because of the q is distributed on the surface, no net charge is inside of sphere. $E_{inside} = 0$



Applying Gauss' Law – Charged Isolated Conductor

Theorem: All excess charges in a charged isolated conductor is placed on the *outer surface* of the conductor.

- Even if there is a cavity inside the conductor, the Gaussian surface shown in figure encloses no net charge. **Why?**
- Therefore, excess charges must be on the outer surface and *not* on the inner surface.
- Electric field inside the cavity is always zero, no matter how big the cavity is (how thin the conductor is).



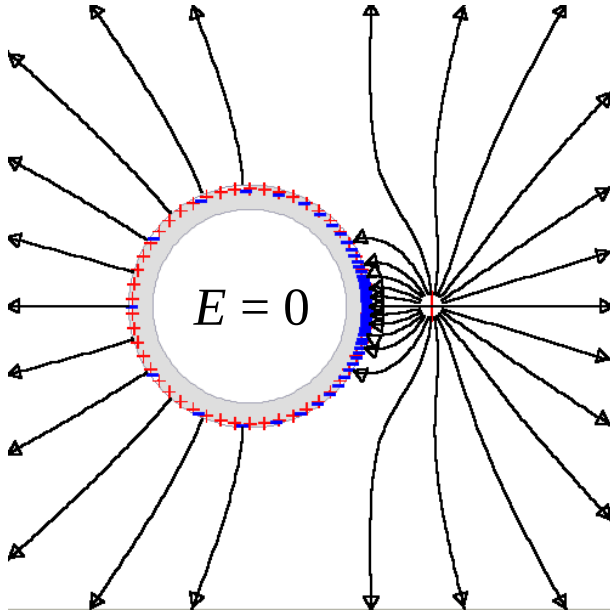
This unique property of metallic conductors can be used for safety purpose: electric shielding (see the next five slides).

Applying Gauss' Law – Electric Shielding

Case 1: Excess charge q **outside** an isolated metallic shell

The field inside the shell is always zero (last slide), so a conductor can be used to shield external electric field.

Will the field penetrate the grid??



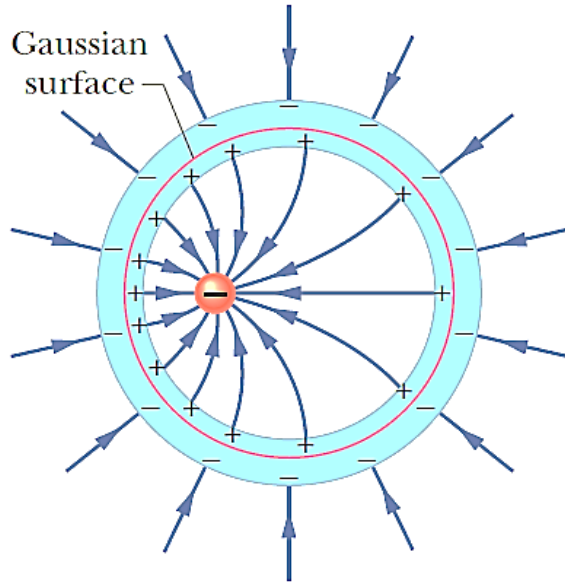
It is more than just playing cool...



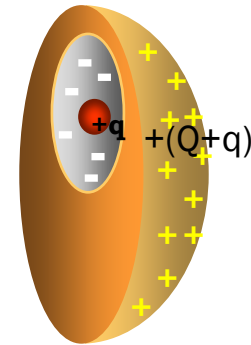
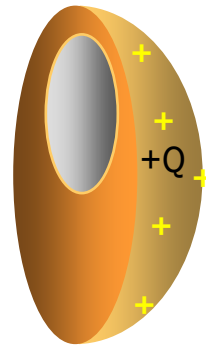
A metallic cage shields a D-Wave quantum computer!

Applying Gauss' Law – Electric Shielding

Case 2: Excess charge $-q$ **inside** an isolated metallic shell



Construct a Gaussian surface inside the conductor. Since $E = 0$ in the conductor, excess charges on the inner surface of the shell must equal to $+q$, and excess charge on the outer surface must equal to $-q$ (why?). So the field outside the shell is the same as an isolated conductor of the same shape with excess charge $-q$.

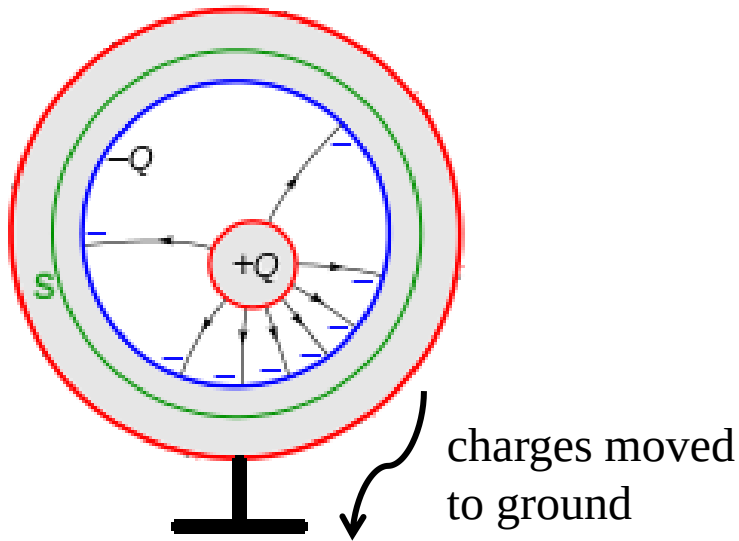


How does the outside electric field change when the charge moves inside? Why?

Applying Gauss' Law – Electric Shielding

Case 3: Excess charge Q inside a *grounded* metallic shell

Grounding of the conductor links it to the *conductive* earth. So the positive charges on the outer surface in Case 2 now distributed all around the surface of the earth. Therefore the outer surface of our small conductor contains zero charge. $q_{\text{outer}} = 0$, $E_{\text{outer}} = 0$.

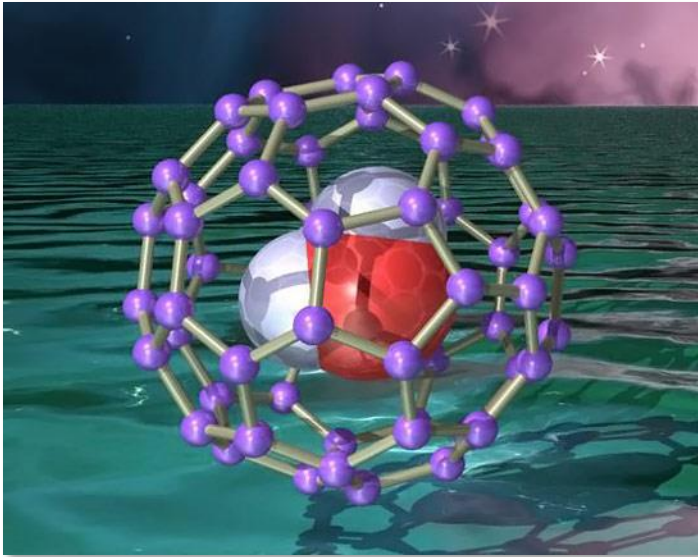


Metallic cages not only shield outside electric field, but also confine the electric field generated by inside charges within the cages (when grounded).

This feature can be also used for safety purpose: electrical instruments usually have a grounded outside metallic case.

Applying Gauss' Law – Electric Shielding

Endohedral Fullerene

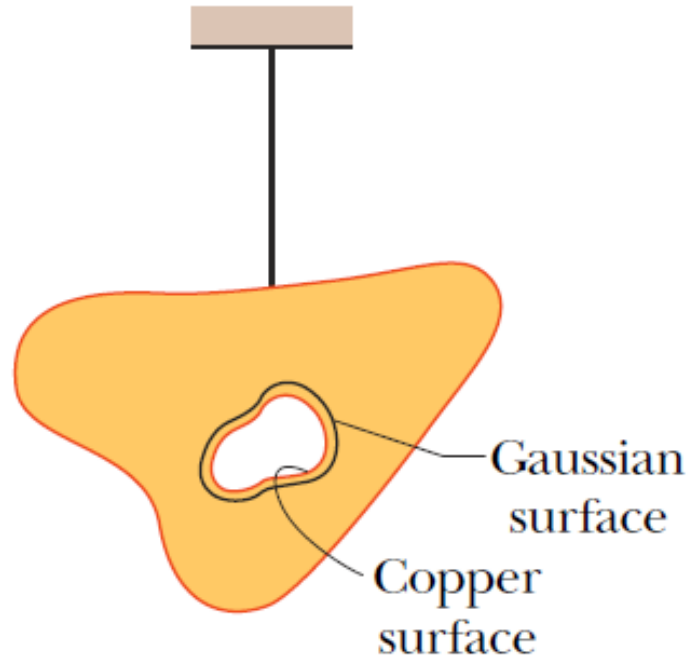


A **fullerene** is a molecule of carbon (C) in the form of a hollow sphere. In particular, molecules of C_{60} resembles a soccer ball. It is discovered in 2011 that a single water molecule (H_2O) can be trapped into the inner of C_{60} , creating what's called an **water-endohedral fullerene**.

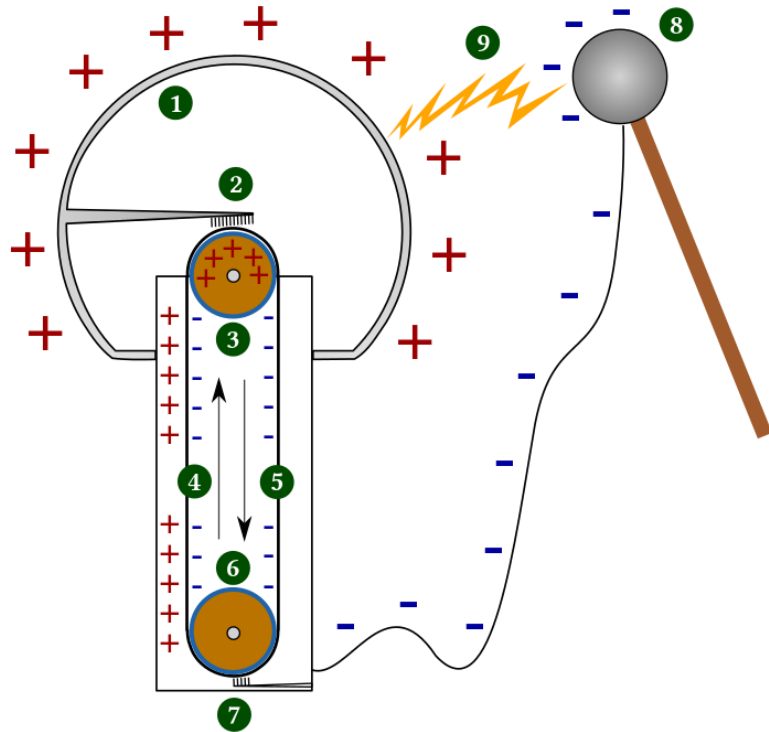
In first-order approximation, the carbon layer in $H_2O@C_{60}$ can be treated as a thin uniform shell of conductor. Using this assumption, estimate the electric field inside and outside the $H_2O@C_{60}$ system but not very close to any atom.

Applying Gauss' Law – Brain Storm

Total charge in Universe = ????



Van de Graaff Generator



1. hollow metal sphere
2. upper electrode
3. upper roller (for example an acrylic glass)
4. side of the belt with positive charges
5. opposite side of belt, with negative charges
6. lower roller (metal)
7. lower electrode (ground)
8. spherical device with negative charges
9. spark produced by the difference of potentials

Robert J. Van de Graaff



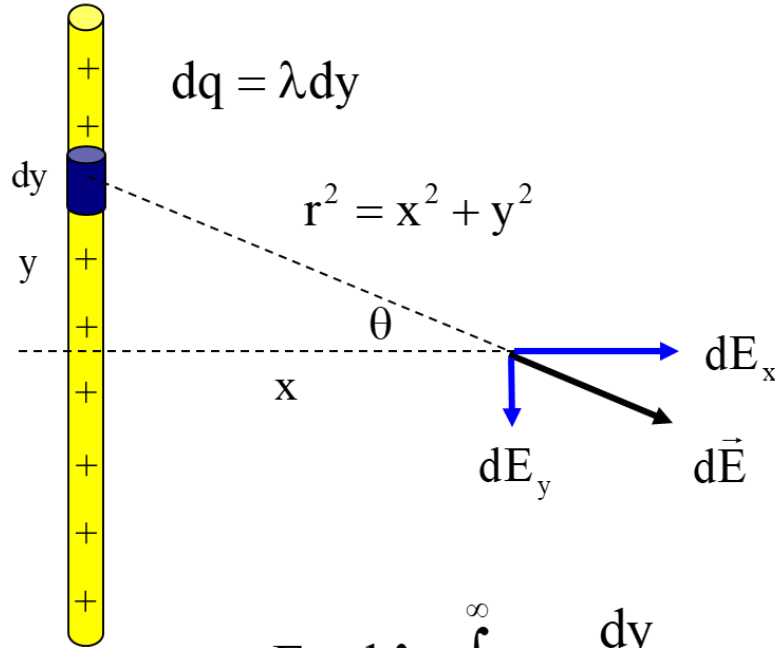
Combine Van de Graaff Generator and Electric Shielding



Today's topics

- Electric Flux – Pre Gauss's law
- Gauss' Law
- Electric Shielding - Metal
- Various Symmetries - Gauss's law applied

Re-cap



$$d\vec{E} = \frac{k dq}{r^2} \hat{r}$$

y-components cancel by symmetry

$$dE_x = \frac{k dq}{r^2} \cos \theta$$

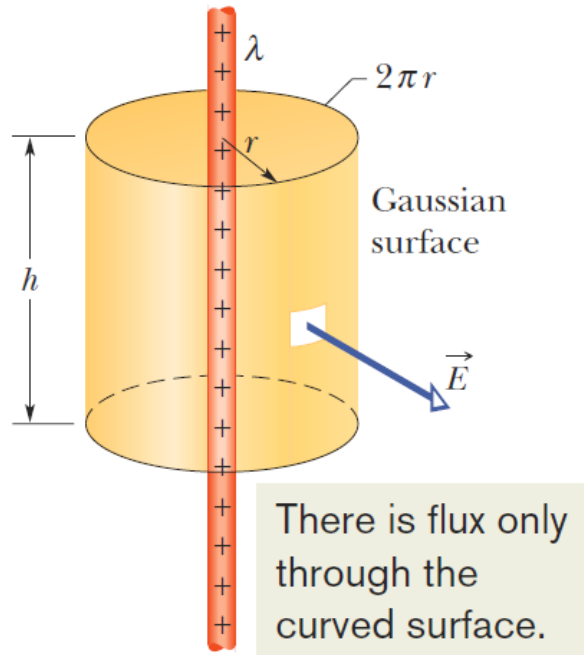
$$dE = \frac{k(\lambda dy)}{x^2 + y^2} \frac{x}{\sqrt{x^2 + y^2}}$$

$$E = k\lambda x \int_{-\infty}^{\infty} \frac{dy}{(x^2 + y^2)^{\frac{3}{2}}} = k\lambda x \left(\frac{2}{x^2} \right) = \frac{2k\lambda}{x}$$

Inversely proportional to the distance!

Applying Gauss' Law – Cylindrical Symmetry

Gauss' Law $\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{enc}}}{\epsilon_0}$



Problem: Find the electric field generated by an infinitely long rod (straight line) with uniform charge density λ .

Try to solve it by integrating contribution of the whole rod

Construct the Gaussian surface as shown. Clearly the top and bottom surfaces contains zero flux (field parallel to surface). On the side surface, E is constant and parallel to surface normal, so

$$\mathbf{E} \cdot d\mathbf{A} = E dA.$$

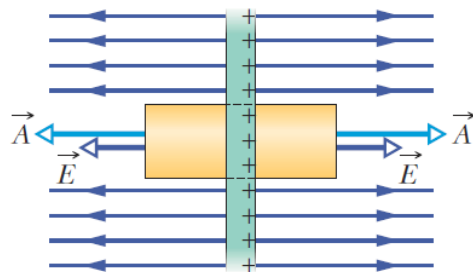
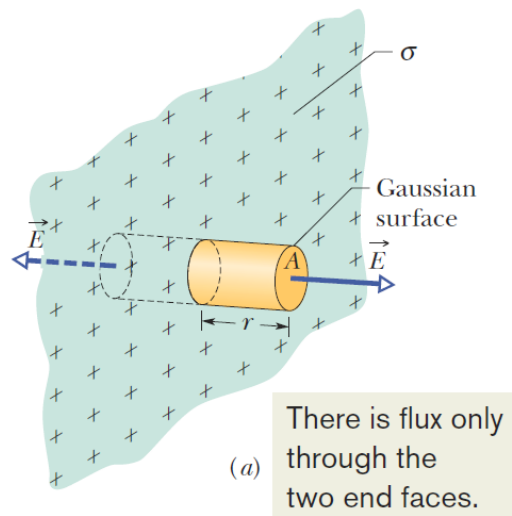
$$\oint \mathbf{E} \cdot d\mathbf{A} = E \int_{\text{side}} dA = E 2\pi r h = \frac{q_{\text{enc}}}{\epsilon_0} = \lambda h / \epsilon_0$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r}.$$

Note that h does not appear in the final result.

Applying Gauss' Law – Planar Symmetry

Gauss' Law $\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{enc}}}{\epsilon_0}$



Problem: Find the electric field generated by an infinite plane with uniform charge density σ .

Try to solve it by integrating contribution of the whole plane

Construct the Gaussian surface as shown (**A does not need to be a circle**). Clearly the side surface contains zero flux (field parallel to surface). On the top and bottom surface, E is constant and parallel to surface normal, so

$$\mathbf{E} \cdot d\mathbf{A} = E dA.$$

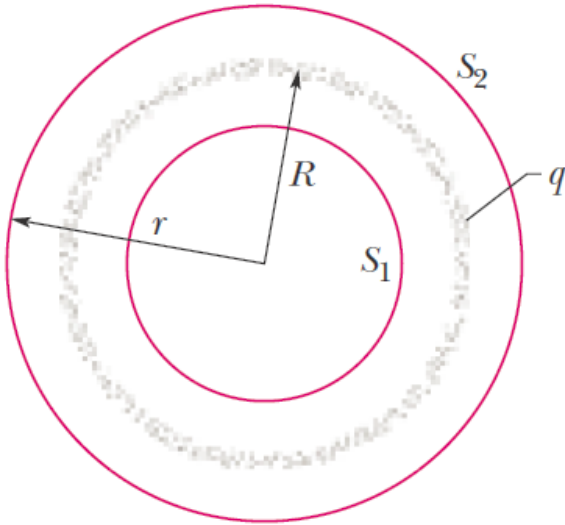
$$\oint \mathbf{E} \cdot d\mathbf{A} = 2E \int_{\text{top}} dA = 2EA = \frac{q_{\text{enc}}}{\epsilon_0} = \frac{\sigma A}{\epsilon_0}$$

$$E = \frac{\sigma}{2\epsilon_0}.$$

This field does not decay with r .

Applying Gauss' Law – Spherical Symmetry

Gauss' Law $\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{enc}}}{\epsilon_0}$



Shell Theorem 1: A *spherical* shell of uniform charge generates the same force to a charged particle outside the shell as if all charges of the shell are concentrated at the center.

Shell Theorem 2: A *spherical* shell of uniform charge generates *no force* on a charged particle inside the shell.

Proof: For a charge outside the shell, we choose S_2 as the G-surface. Apparently in this surface. So

$$\mathbf{E} \cdot d\mathbf{A} = E dA$$

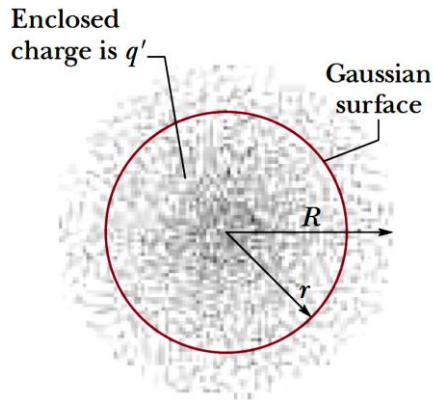
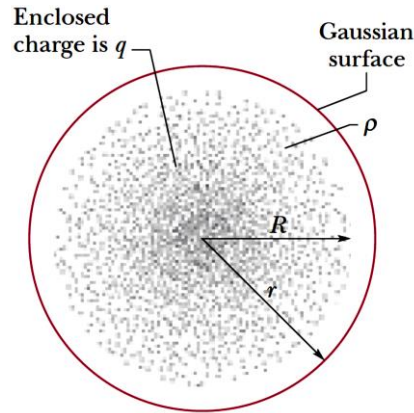
$$\oint \mathbf{E} \cdot d\mathbf{A} = E \int_{S_2} dA = E 4\pi r^2 = \frac{q}{\epsilon_0}, \quad E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}.$$

For a charge inside the shell, we choose S_1 as the G-surface.

$$\oint \mathbf{E} \cdot d\mathbf{A} = E \int_{S_1} dA = E 4\pi r^2 = 0 \Rightarrow E = 0.$$

Applying Gauss' Law – Spherical Symmetry

Gauss' Law $\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{enc}}}{\epsilon_0}$



Problem: Find the electric field generated by a sphere of radius R , whose charge distributes uniformly throughout the *volume*.

1. $r > R$.

Construct the Gaussian surface as shown. Clearly it contains all charges:

$$q_{\text{enc}} = Q = \rho V = \frac{4}{3}\rho\pi R^3. \quad \oint \mathbf{E} \cdot d\mathbf{A} = E \oint dA = \frac{q_{\text{enc}}}{\epsilon_0},$$

$$E4\pi r^2 = \frac{q_{\text{enc}}}{\epsilon_0} = \frac{4\rho\pi R^3}{3\epsilon_0}, \quad E = \frac{\rho R^3}{3\epsilon_0 r^2}.$$

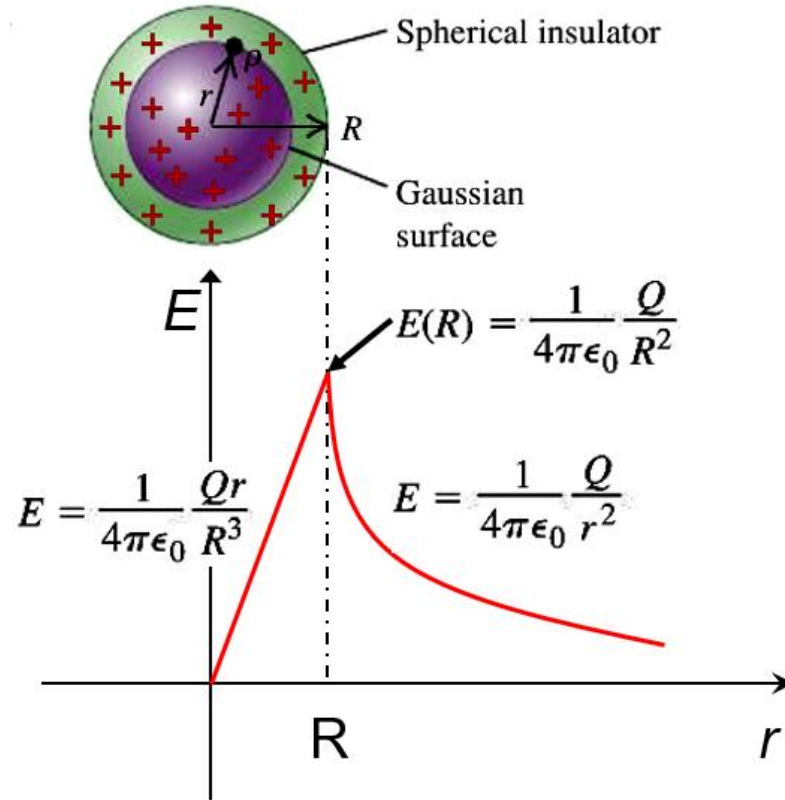
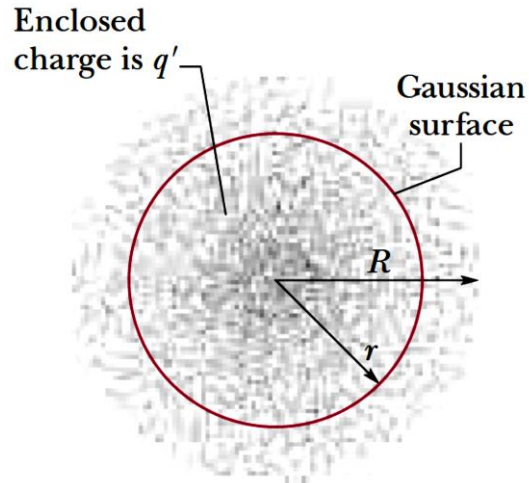
2. $r < R$.

Construct the Gaussian surface as shown. It now contains part of the charges:

$$q_{\text{enc}} = \rho V(r) = \frac{4}{3}\rho\pi r^3. \quad \oint \mathbf{E} \cdot d\mathbf{A} = E \oint dA = \frac{q_{\text{enc}}}{\epsilon_0},$$

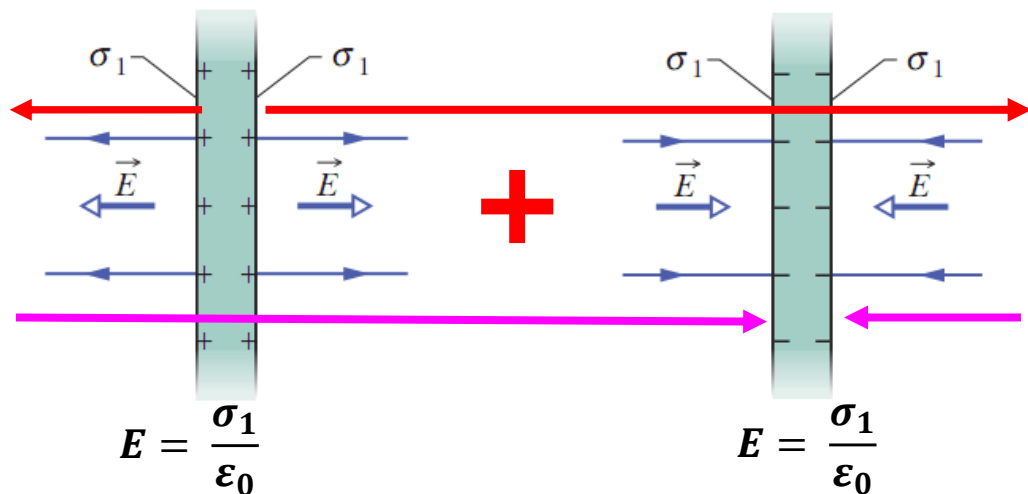
$$E4\pi r^2 = \frac{q_{\text{enc}}}{\epsilon_0} = \frac{4\rho\pi r^3}{3\epsilon_0}, \quad E = \frac{\rho r}{3\epsilon_0}.$$

Applying Gauss' Law – Spherical Symmetry

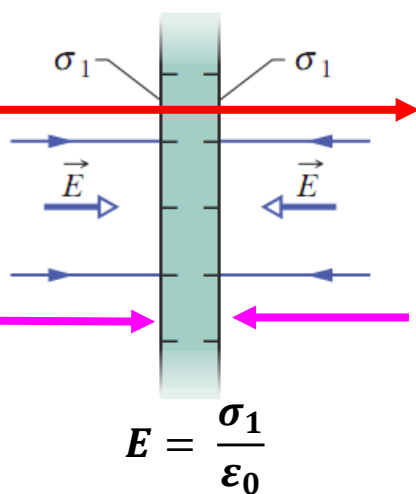


Applying Gauss' Law – Conducting Plates

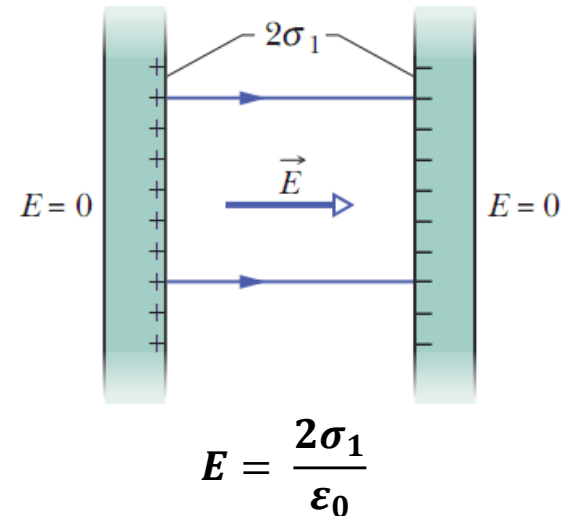
Thin, **infinite conducting** plate with excess charge with surface charge density, σ_1 . Find the electric field outside of the plate.



Directed away the plate



Directed toward the plate



From positive to negative plate

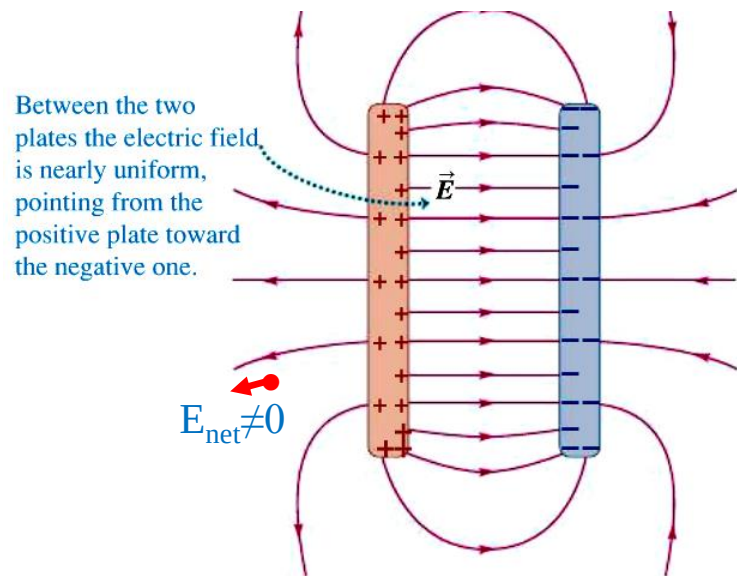
$$E = \frac{2\sigma_1}{\epsilon_0}$$



holds well for a pair of finite conducting plates for the points that are not too close to the edges.

Two differently charged conducting plates

Applying Gauss' Law – Planar symmetry

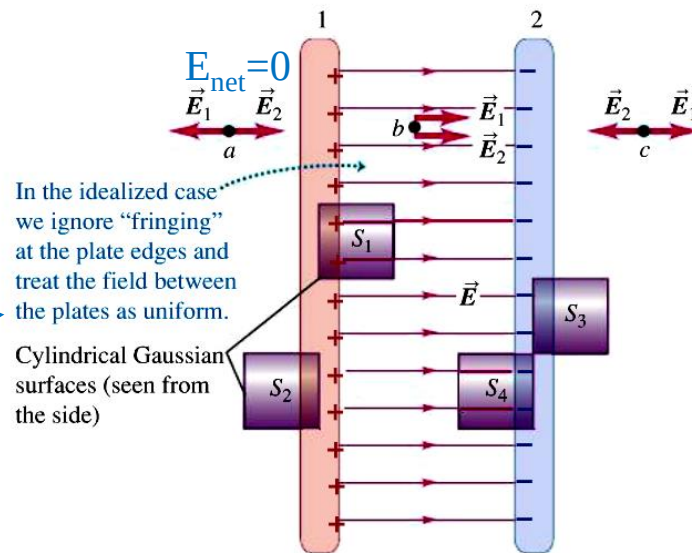


Nonconducting plates

$$EA = \sigma A / \epsilon_0$$

$$E = \frac{\sigma}{\epsilon_0} \text{ (field between oppositely charged conducting plates)}$$

We will use this approximation



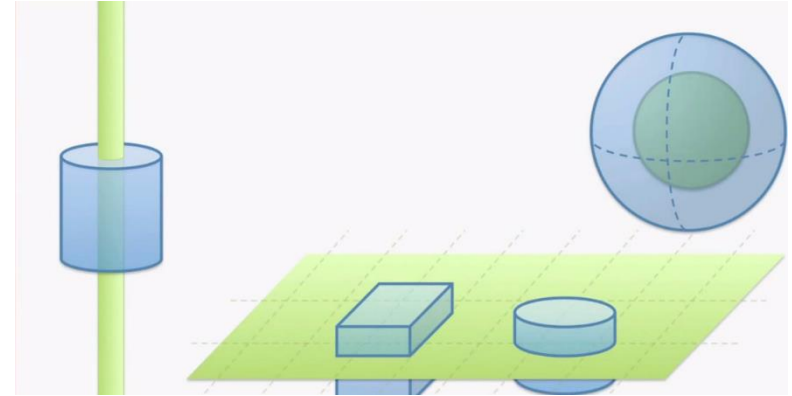
Conducting plates

Where Can Gauss' Law Be Applied?

Charge distribution that Gauss' Law will NOT Help



Some examples that Gauss' Law applies



Gauss' law makes it possible to find the electric field easily in **highly symmetric situations**, but you have to understand its physics!

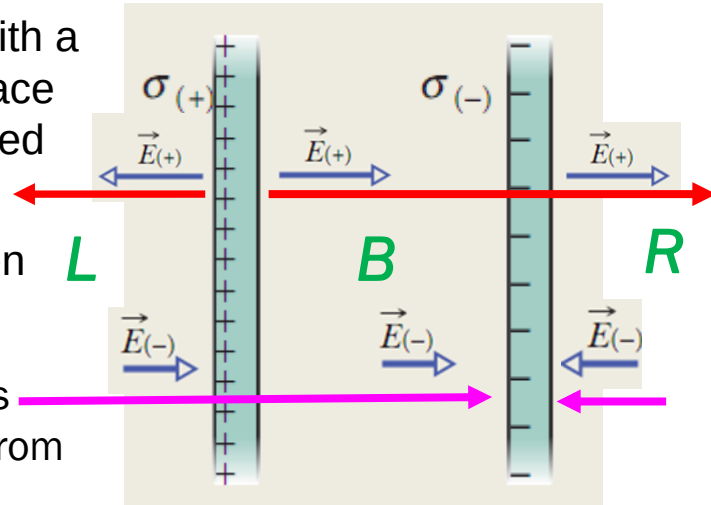


Sample Problem

Portions of two large, parallel, non-conducting sheets, each with a fixed uniform charge on one side. The magnitudes of the surface charge densities are $\sigma_{(+)} = 6.8 \mu\text{C}/\text{m}^2$ for the positively charged sheet and $\sigma_{(-)} = 4.3 \mu\text{C}/\text{m}^2$ for the negatively charged sheet.

Find the electric field $\vec{E}$ (a) to the left of the sheets, (b) between the sheets, and (c) to the right of the sheets.

Solution: At any point, the electric field due to the positive sheet is directed away and due to the negative sheet is directed towards from the sheets, we have



$$E_{(+)} = \frac{\sigma_{(+)}}{2\epsilon_0} = \frac{6.8 \times 10^{-6} \text{ C/m}^2}{(2)(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} = 3.84 \times 10^5 \text{ N/C}.$$

$$E_{(-)} = \frac{\sigma_{(-)}}{2\epsilon_0} = \frac{4.3 \times 10^{-6} \text{ C/m}^2}{(2)(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} = 2.43 \times 10^5 \text{ N/C}.$$

$$\begin{aligned} E_L &= E_{(+)} - E_{(-)} \\ &= 3.84 \times 10^5 \text{ N/C} - 2.43 \times 10^5 \text{ N/C} \\ &= 1.4 \times 10^5 \text{ N/C}. \end{aligned}$$

$$\begin{aligned} E_B &= E_{(+)} + E_{(-)} \\ &= 3.84 \times 10^5 \text{ N/C} + 2.43 \times 10^5 \text{ N/C} \\ &= 6.3 \times 10^5 \text{ N/C}. \end{aligned}$$



Sample Problem

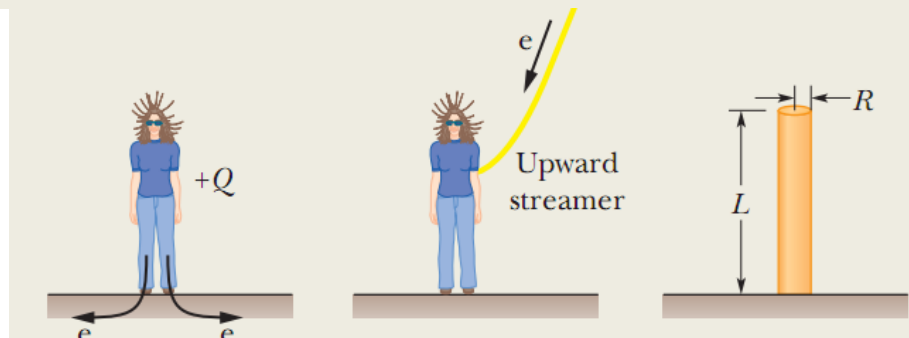
Upward streamer in a lightning storm. The woman in Fig. 23-15 was standing on a lookout platform high in the Sequoia National Park when a large storm cloud moved overhead. Some of the conduction electrons in her body were driven into the ground by the cloud's negatively charged base (Fig. 23-16a), leaving her positively charged. You can tell she was highly charged because her hair strands repelled one another and extended away from her along the electric field lines produced by the charge on her.



Courtesy NOAA

Let's model her body as a narrow vertical cylinder of height $L = 1.8$ m and radius $R = 0.10$ m (Fig. 23-16c). Assume that charge Q was uniformly distributed along the cylinder and that electrical breakdown would have occurred if the electric

field magnitude along her body had exceeded the critical value $E_c = 2.4$ MN/C. What value of Q would have put the air along her body on the verge of breakdown?



Calculations: Substituting the critical value E_c for E , the cylinder radius R for radial distance r , and the ratio Q/L for linear charge density λ , we have

$$E_c = \frac{Q/L}{2\pi\epsilon_0 R},$$

or

$$Q = 2\pi\epsilon_0 R L E_c.$$

Substituting given data then gives us

$$\begin{aligned} Q &= (2\pi)(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(0.10 \text{ m}) \\ &\quad \times (1.8 \text{ m})(2.4 \times 10^6 \text{ N/C}) \\ &= 2.402 \times 10^{-5} \text{ C} \approx 24 \mu\text{C}. \end{aligned}$$

(Answer)

Gauss' Law Gauss' law and Coulomb's law are different ways of describing the relation between charge and electric field in static situations. Gauss' law is

$$\epsilon_0 \Phi = q_{\text{enc}} \quad (\text{Gauss' law}), \quad (23-6)$$

in which q_{enc} is the net charge inside an imaginary closed surface (a *Gaussian surface*) and Φ is the net *flux* of the electric field through the surface:

$$\Phi = \oint \vec{E} \cdot d\vec{A} \quad (\text{electric flux through a Gaussian surface}). \quad (23-4)$$

Coulomb's law can be derived from Gauss' law.

Applications of Gauss' Law Using Gauss' law and, in some cases, symmetry arguments, we can derive several important results in electrostatic situations. Among these are:

1. An excess charge on an isolated *conductor* is located entirely on the outer surface of the conductor.
2. The external electric field near the *surface of a charged conductor* is perpendicular to the surface and has a magnitude that depends on the surface charge density σ :

$$E = \frac{\sigma}{\epsilon_0} \quad (\text{conducting surface}). \quad (23-11)$$

Within the conductor, $E = 0$.

3. The electric field at any point due to an infinite *line of charge*

with uniform linear charge density λ is perpendicular to the line of charge and has magnitude

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \quad (\text{line of charge}), \quad (23-12)$$

where r is the perpendicular distance from the line of charge to the point.

4. The electric field due to an *infinite nonconducting sheet* with uniform surface charge density σ is perpendicular to the plane of the sheet and has magnitude

$$E = \frac{\sigma}{2\epsilon_0} \quad (\text{sheet of charge}). \quad (23-13)$$

5. The electric field *outside a spherical shell of charge* with radius R and total charge q is directed radially and has magnitude

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \quad (\text{spherical shell, for } r \geq R). \quad (23-15)$$

Here r is the distance from the center of the shell to the point at which E is measured. (The charge behaves, for external points, as if it were all located at the center of the sphere.) The field *inside* a uniform spherical shell of charge is exactly zero:

$$E = 0 \quad (\text{spherical shell, for } r < R). \quad (23-16)$$

6. The electric field *inside a uniform sphere of charge* is directed radially and has magnitude

$$E = \left(\frac{q}{4\pi\epsilon_0 R^3} \right) r. \quad (23-20)$$